Video Transcript

Video 1: Introduction (05:53)

So, this week we're going to do an overview of machine learning methods. We don't have time to go into great depth, but we're going to cover some of the techniques that are used to learn. So, there are two kinds of machine learning techniques, supervised learning, and unsupervised learning. We're going to look at supervised learning, where we have a training set of data where we know the answers. And we're going to build a model, and now our model is going to be linear regression. So, we're going to put a straight line, if you like, through a number of points.

Now there's going to be some error between our model and this training data, and our goal is to adjust the parameters of the model so that we minimize the error. Now, there are various metrics of error, we're going to use least squares. Some of the others involve cross entropy and other metrics that can be used. But least squares is an interesting one in that it allows us to use an iterative method. And in machine learning, we almost always iterating so that we're trying to minimize this error and get to a good solution, build a good model.

Now, we saw in linear regression that we need to choose features that are correlated with the parameter that we're trying to predict. So, feature selection is essential part and you've already covered in the data science part that we've done in this course, how to choose features based on correlation coefficients. Now we're going to look at how we minimize this error. We're choosing least squares, and least squares has an interesting property in that the error is shaped like a bowl. It's a parabola. So, the further we are away from the solution, the higher up the sides of this parabola we are.

Now initially, we're going to have quite a large error and so we're way up the sides of this parabola and we need to get to the minimum where the error is a minimum. So, we need to come down this bowl and what we do is we find the gradient that point that we're at and we follow that gradient. Now, the problem is as we move down this bowl, the gradient changes. So, we don't want to move too far before we recalculate the gradient. So, some of the tricks, if you like of machine learning are to choose how far we go in a step, how far down do we go? Because

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if we choose a gradient, it's a straight line, but the parabola is curved. So, we're going to move off our parabola, if you like.

Now once we get close to the solution as well, there's a chance that we're going to overshoot and our error is going to increase. So, we need to monitor the error. And then as we get close to the solution, we're probably going to change the size of our steps and make them smaller and smaller so that we don't overshoot. So, gradient descent, and then we're going to look at Bayes, Naive Bayes sneak, which is called Naive Bayes. Now Bayes's theorem basically relates how two sets intersect. So, we've got set A and set B and they intersect. Now that intersection can be defined two ways. It can be defined as given that we're in A, then the probability that we're also in B. Or we can define it as, given that we're in B, the probability that we're also in A. Now, those two statements need to be equal, and that's what Bayes's theorem does. It takes advantage of exactly that, that those two statements are equal. Now, the way we're going to apply it is going to be to calculating how many rich people are actually happy.

And we're going to be going to be given the information about how many people in the population are happy and what proportion of those happy people are rich. And it's a small proportion, but what we're after is how many rich people are happy. So, you're going to apply Bayes theorem to calculate that, and you'll often hear it as given the prior probability, we calculate a posterior probability based on some other data that we're given. And we'll then going to apply Naive Bayes to spam messages, understanding which messages are spam and which are not. And then also to another problem where we're going to use an assumption that our data are about our features if you like, our data is Gaussian and we're going to use probability density functions as measures of the probabilities that we're going to plug into Naive Bayes. So, let's take a look now at starting on machine learning.

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